Colin Friedel

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EDUCATION

Santa Clara University, Santa Clara, CA June 2026

B.S. Computer Science and Engineering

Certifications: AI Fluency Framework & Foundations (Anthropic, June 2026) · Claude 101 (Anthropic, July 2026)

Relevant Coursework: Software Engineering, Operating Systems, AI, Algorithms, Computer Networks, Embedded Systems

PROJECTS

Reef Monitoring Communication System

C, RS-485, RS-232, Acoustic Modems, Cellular | github.com/colinfriedel/IRIS-Senior-Design

- Led the software and data pipeline for an underwater reef sensor system designed for the Palos Verdes Reef, coordinating cellular, acoustic, and serial communication from sensor to cloud, and validated it through pool and harbor testing
- Wired and programmed an RS-485 interface to buffer sensor data on a Feather Adalogger M0, then converted it to RS-232 to drive a Succorfish Delphis acoustic modem that transmitted it through the water to a matching receiving chain
- Programmed a Particle Boron 4G LTE module (404X) to publish the data to a Google Sheet on a public Google Site

Dipstick: Car Maintenance and Mileage Tracker

Swift, SwiftUI, SwiftData | github.com/colinfriedel/Dipstick

- Developing a native iPhone app that tracks car maintenance and shows real fuel efficiency split into highway and city, helping users spot possible problems
- Records speed whenever the phone connects to the car over Bluetooth or CarPlay to separate highway and city mileage, with all data stored on the device in SwiftData
- Includes logs, charts, and maintenance reminders; testing the tracking through regular personal use

Car Stereo Controller

Arduino Nano, C++ | github.com/colinfriedel/jvc-carplay-remote

- Built a physical volume knob for an Acura RSX by wiring a repurposed sunroof switch to an Arduino Nano that commands the JVC head unit through its steering wheel remote input
- Mapped press, hold, and double-tap gestures on one switch to volume, track skip, mute, and voice assistant, using command codes found through testing
- Soldered and multimeter-tested the connections, and powered the board from the car's 12V line through a buck converter

Automatic Radio Recording Distribution System

Python, AWS EC2, AWS S3 | github.com/colinfriedel/KSCU-Show-Recorder

- Built a continuously running Python service on AWS EC2 that auto-records hourly radio output and emails each recording directly to the on-air DJ in a downloadable format
- Architected a cloud storage pipeline using AWS S3; runs unattended 24/7 with zero manual intervention

WORK EXPERIENCE

General Manager, KSCU 103.3 FM Radio Station | Santa Clara, CA June 2025 - June 2026

- Managed 16 hired student staff across programming, production, and operations, and oversaw 130+ active DJs and volunteers to sustain a 24/7 live broadcast
- Planned and executed multiple live music events per quarter, coordinating artists, venues, and logistics end-to-end
- Was solely responsible for staff hiring, station strategy, and day-to-day operations

Software Development Intern, NationsBenefits, LLC | Plantation, FL June - August 2023

- Developed and optimized C# programs to parse and reformat healthcare data files received from external partners, improving processing efficiency
- Operated in an Agile/Scrum environment; presented technical progress and findings to the software development team

SKILLS

Languages: Python, C, C++, C#, JavaScript, Swift, HTML, CSS

Tools & Platforms: AWS, React, Git/GitHub, VS Code, Xcode, Arduino IDE, Azure DevOps, Jira

AI Tools: Claude chat, Projects, and Claude Code, used in developing Dipstick, the Car Stereo Controller, and my portfolio site

Hardware: Arduino, Particle Boron, RS-485/RS-232, soldering, multimeter testing

Concepts: Full-stack development, REST APIs, Agile/Scrum, cloud infrastructure, embedded systems

ADDITIONAL

Eagle Scout (2022) · Backpacking · Rock Climbing · Automotive Repair · Guitar (monthly volunteer performances) · Trivia